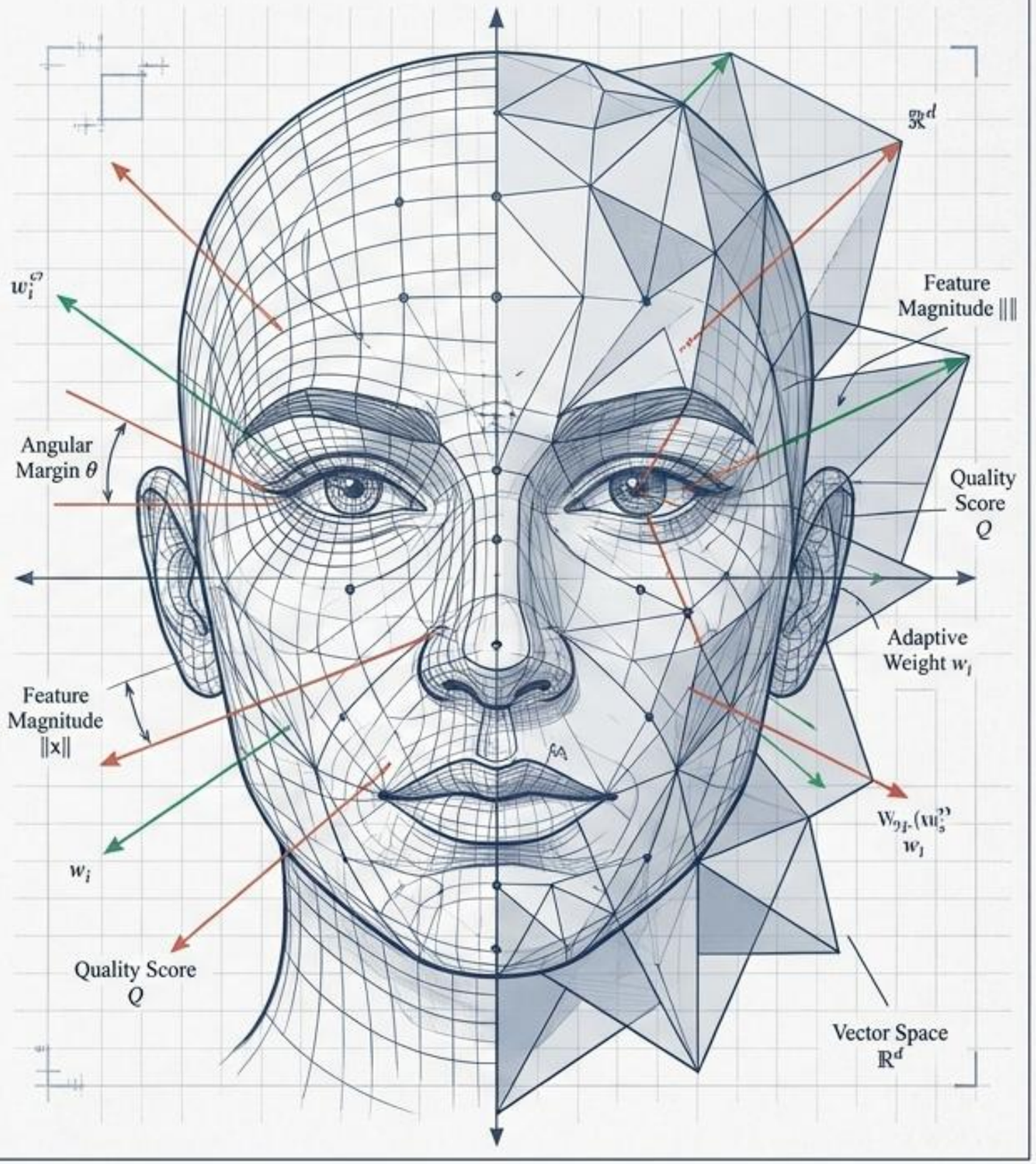
Sardar Vallabhbhai National Institute of Technology (SVNIT), Surat
Department of Artificial Intelligence
Course: Advanced Biometric Systems and Security (AI461)
Phase I – Literature Review & State-of-the-Art

Deep Face Recognition: From Adaptive Curriculum Learning to Quality-Aware Representations

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Project Type:	Individual Project
Professor:	Dr. Praveen Kumar Chandaliya



The Fundamentals of Biometric Face Recognition

Biometric Authentication: Utilizing physiological traits to verify identity securely.



Verification (1:1)

Confirming if a person is who they claim to be.



Identification (1:N)

Searching a vast database to determine a subject's identity.



The Core Challenge – Open-Set Recognition

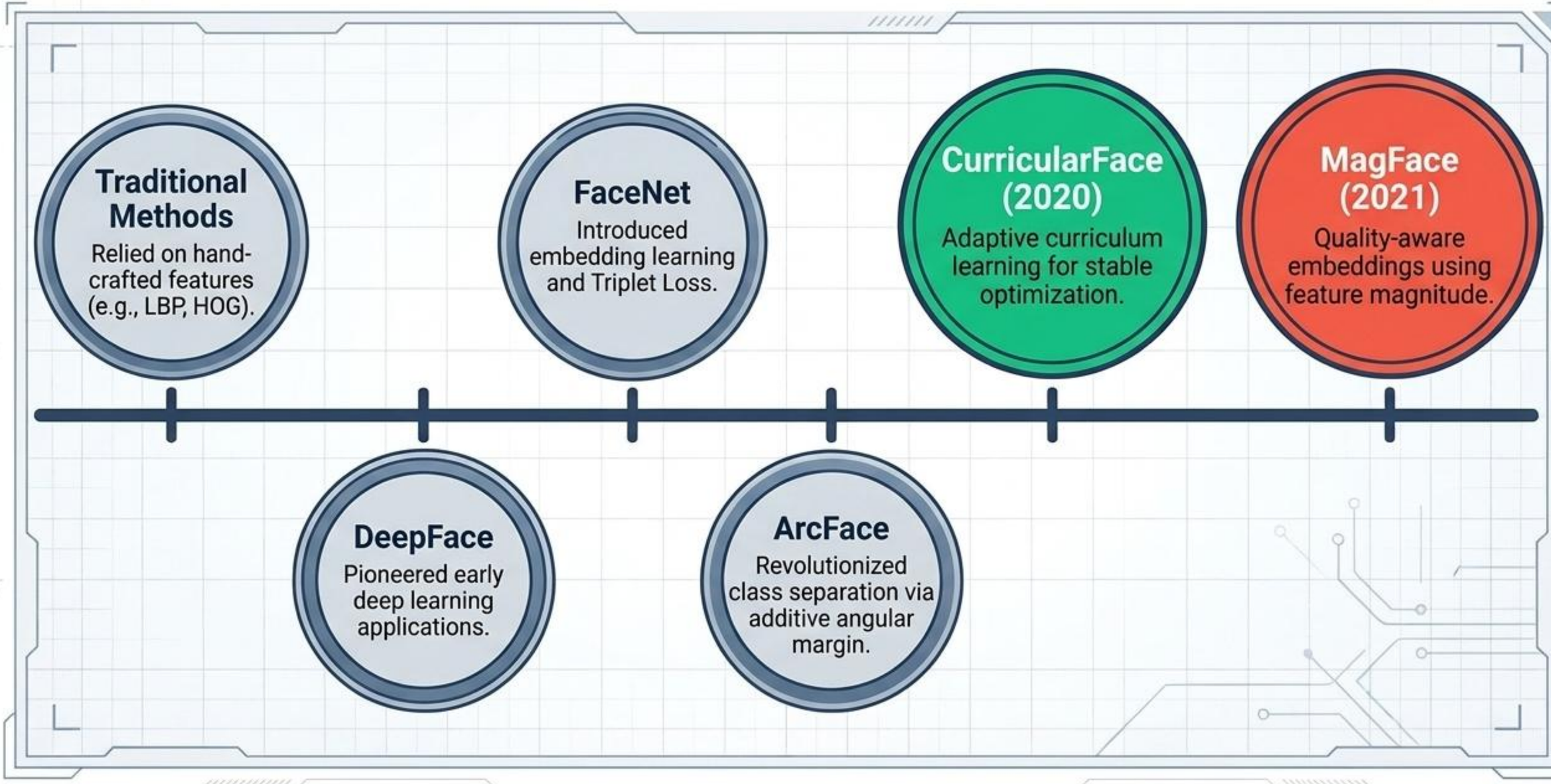
Practical scenarios where testing identities are entirely disjoint from the training set, requiring highly discriminative feature learning rather than mere memorization.



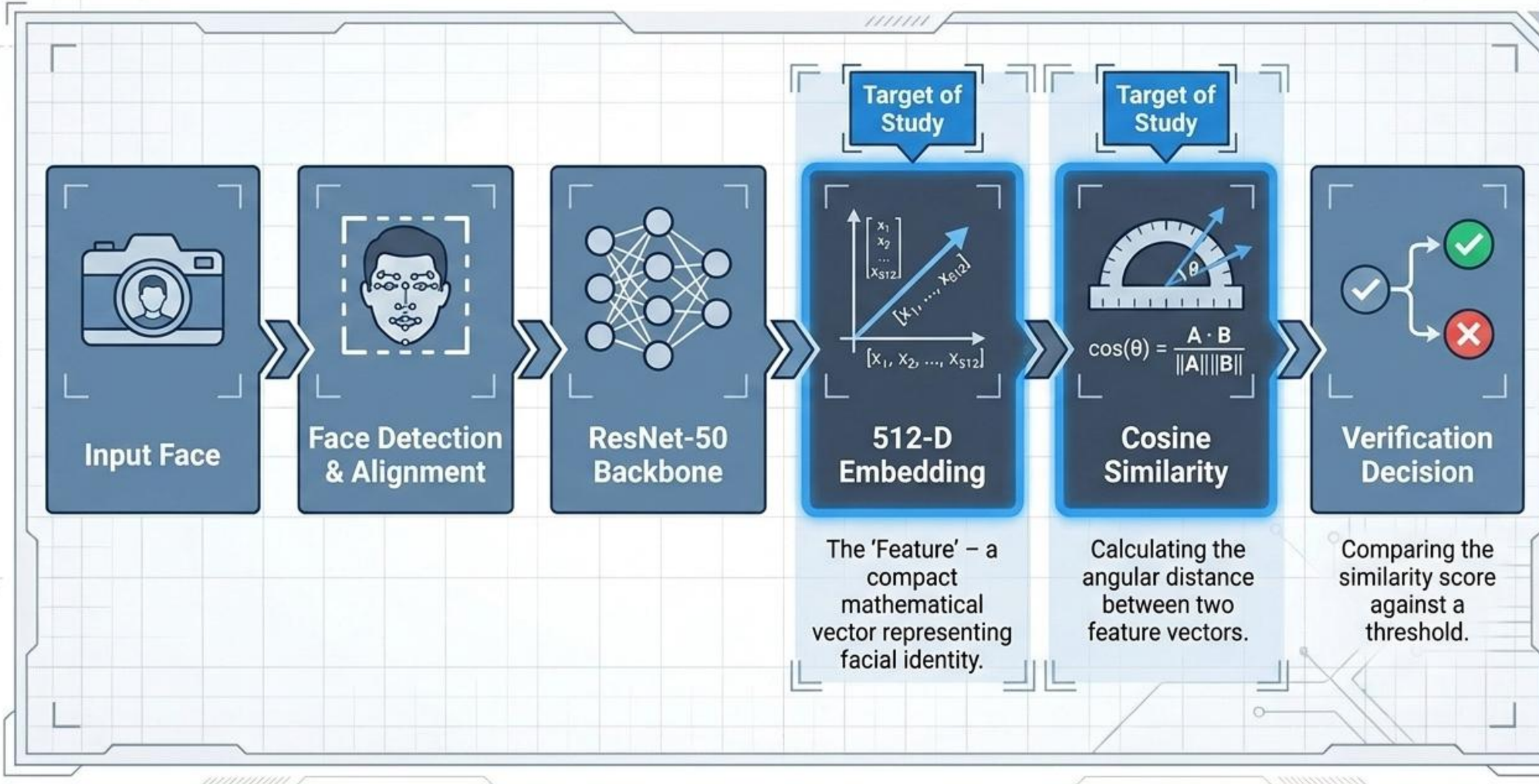
Real-World Variability

Handling extreme variations in lighting, pose, and blur.

The Evolution of Deep Face Recognition



The Deep Face Recognition Pipeline



CurricularFace: Motivation & Concept (CVPR 2020)

The Research Gap

Prior margin-based losses (ArcFace) ignore sample difficulty; early mining-based losses cause convergence failure by focusing on hard samples too soon.

Sample Categorization

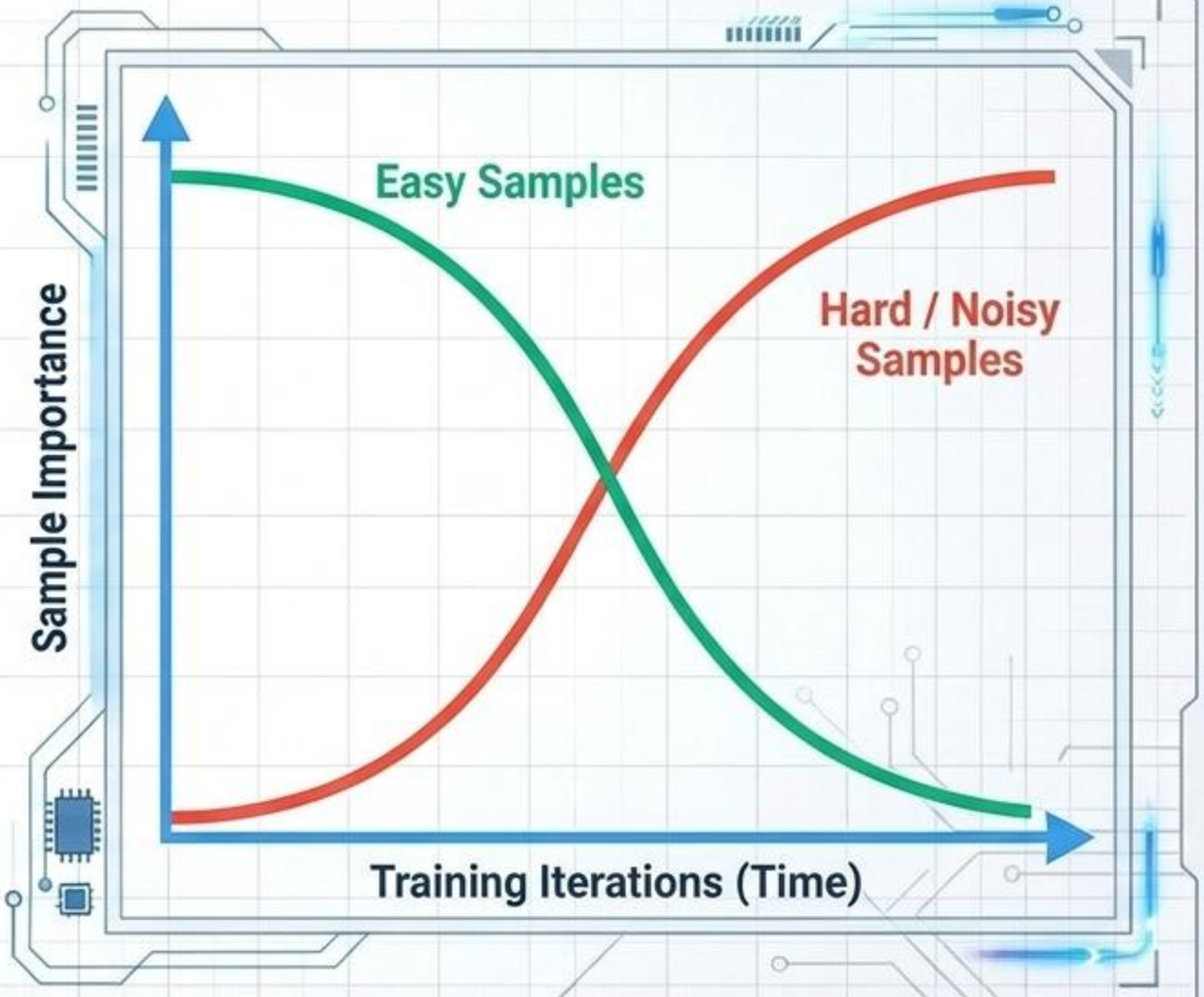
Training data inherently contains easy, semi-hard, and hard (noisy/misclassified) samples.

Curriculum Learning (CL)

Inspired by human cognitive learning—master easy concepts first to build a foundation, tackle hard ones later.

Adaptive Strategy

Dynamically adjusts the importance of samples as training stages progress.



CurricularFace: Adaptive Training Architecture

CurricularFace Loss:

Modulates negative cosine similarities for misclassified samples.

Adaptive Weighting (t):

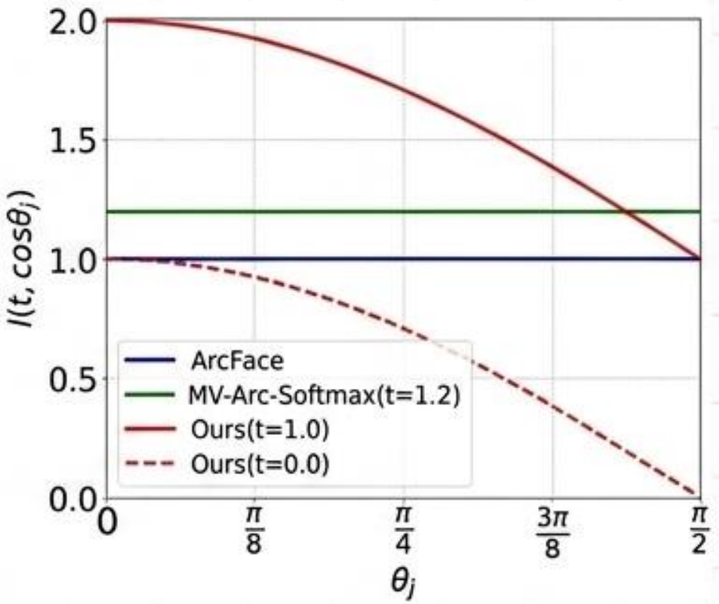
Uses an Exponential Moving Average (EMA) to dynamically estimate the training stage without manual tuning:

$$t^{(k)} = \alpha r^{(k)} + (1 - \alpha) t^{(k-1)}$$

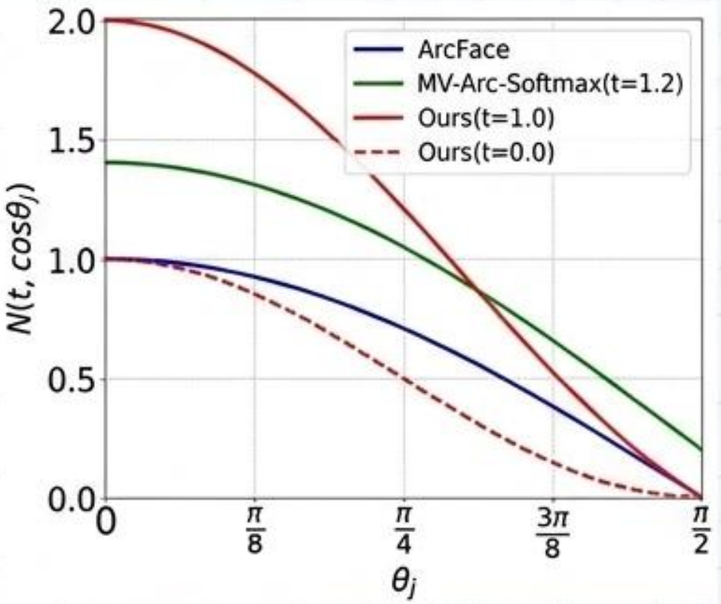
(where $\alpha = 0.99$)

Classification Dynamics:

- **Early Stage (t ≈ 0):** Easy samples dominate loss.
- **Late Stage (t ≈ 1):** Hard samples are mathematically amplified.



Modulation coefficients for negative cosine similarities.



Negative cosine similarities after modulation.



Novel contribution: Curriculum Learning dynamically shifts focus from easy to hard samples

Critique of CurricularFace

Advantages

- Highly stable optimization; avoids early-stage convergence failure.
- Yields highly discriminative features by fully exploiting hard samples.
- Eliminates manual hyperparameter tuning via EMA tracking.

Limitations

- Features remain purely directional (normalized vectors).
- Lacks intrinsic image quality estimation at inference time.



Critical Synthesis

While it beautifully optimizes the training journey, at inference time, it treats a blurry CCTV crop and a high-res studio portrait as structurally equal in the mathematical vector space.

MagFace: Motivation & Universal Concept (CVPR 2021)



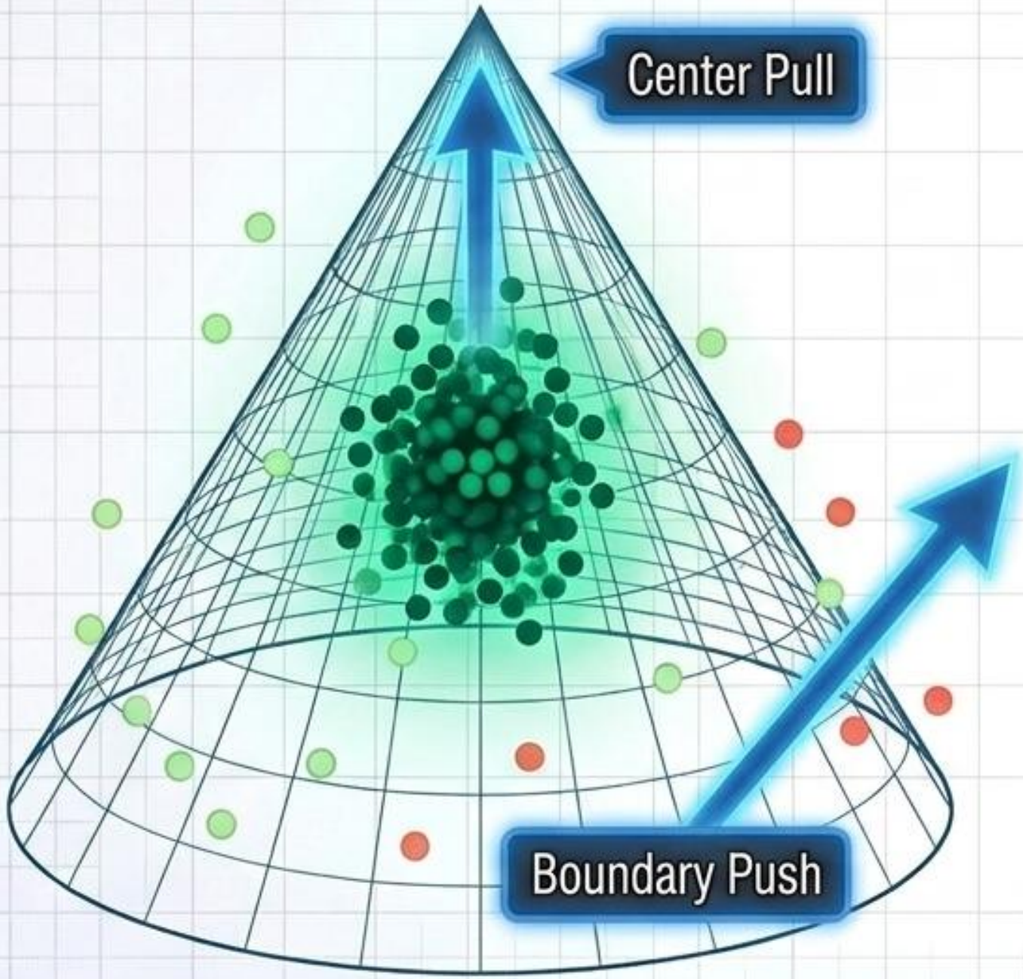
The Universal Representation
One unified feature vector serves two distinct purposes without external labels.

Direction -> Identity
The angular position determines who the person is.

Magnitude -> Quality
The unnormalized length of the vector dictates the quality of the face image.

Dynamic Angular Margin
Pushes ambiguous, low-quality samples away from class centers while pulling high-quality ones inward.

MagFace: Quality-Aware Architecture



Input & Backbone: ResNet-50 extracts raw 512-D unnormalized features.

Magnitude Learning: The loss function specifically rewards vectors with larger magnitudes.

Adaptive Margin (Spatial Structuring):

- **Center Pull:** Easy/High-quality samples (large magnitude) are pulled tightly to the class center.
- **Boundary Push:** Hard/Ambiguous samples (small magnitude) are pushed away toward class boundaries.

Result: A cone-like within-class distribution that isolates noise.



Critique of MagFace



Advantages

- Achieves simultaneous recognition and quality assessment.
- Inherently prevents overfitting on noisy/low-quality samples.
- Requires zero external quality labels during the training process.

Limitations

- Coupling magnitude with angular margin complicates optimization.
- Requires strict mathematical bounding of hyperparameters to prevent magnitude explosion.

Critical Synthesis

MagFace represents a true paradigm shift; by stopping the industry practice of discarding feature magnitude, it elegantly solves two distinct real-world biometric problems with a single vector.

State-of-the-Art Synthesis: CurricularFace vs. MagFace

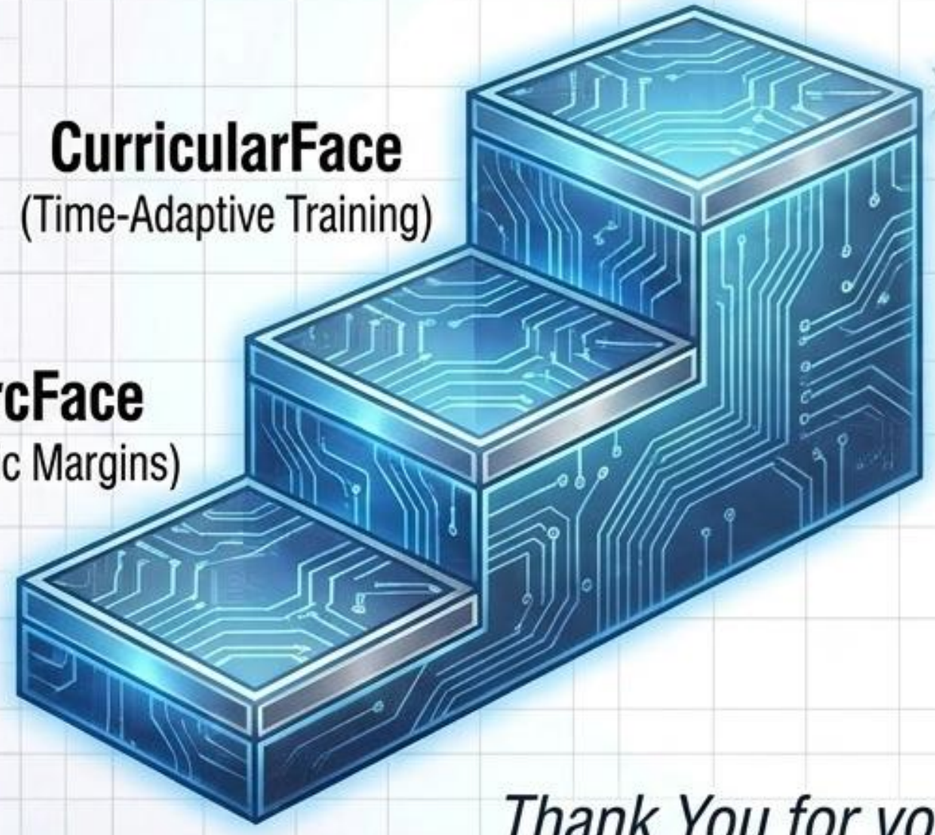
Feature	CurricularFace (2020)	MagFace (2021)
Core Contribution	Adaptive Curriculum Loss	Universal Quality Representation
Feature Focus	Normalized Direction only	Direction + Unnormalized Magnitude
Training Strategy	Time-based (Easy -> Hard)	Spatial-based (Center vs Boundary)
Quality Estimation	None	Built-in via Magnitude
Overfitting Prevention	Dynamic weight modulation	Margin-based boundary pushing
Complexity	Moderate (EMA tracking)	High (Bounded margin functions)

Conclusion & Phase II Objectives

MagFace
(Quality-Aware Embeddings)

CurricularFace
(Time-Adaptive Training)

ArcFace
(Static Margins)



Future Work (Phase II)

- Implement CurricularFace and MagFace in PyTorch.
- Use ResNet-50 with the MS1MV2 dataset.
- Compare recognition performance using Accuracy, ROC, TAR@FAR, and Cosine Similarity.
- Evaluate the quality-aware capability of MagFace against the CurricularFace baseline.

Thank You for your time and attention.